

```

1 # infer_webcam.py
2 # Run locally (needs access to your webcam – won't work in Colab).
3 # pip install ultralytics opencv-python
4 # python infer_webcam.py
5
6 import cv2
7 from ultralytics import YOLO
8
9 # Path to your trained weights (downloaded from runs/ppe/ppe_yolov8n/weights/best.pt)
10 MODEL_PATH = "best.pt"
11 CONFIDENCE_THRESHOLD = 0.5
12
13 model = YOLO(MODEL_PATH)
14
15 cap = cv2.VideoCapture(0) # 0 = default webcam; try 1 if you have multiple cameras
16 if not cap.isOpened():
17     raise RuntimeError("Could not open webcam. Check camera permissions/index.")
18
19 print("Press 'q' to quit.")
20
21 while True:
22     ret, frame = cap.read()
23     if not ret:
24         break
25
26     results = model(frame, conf=CONFIDENCE_THRESHOLD, verbose=False)
27     annotated_frame = results[0].plot() # draws boxes + labels automatically
28
29     cv2.imshow("PPE Compliance Detection", annotated_frame)
30
31     if cv2.waitKey(1) & 0xFF == ord("q"):
32         break
33
34 cap.release()
35 cv2.destroyAllWindows()
36
37 # TIP for your demo GIF/video (for README and LinkedIn):
38 # Screen-record this window running for ~15–20 seconds, trim with any free tool
39 # (e.g. ezgif.com) into a 5–10 second looping GIF under 10MB for GitHub embedding.
40

```

Creating new Ultralytics Settings v0.0.8 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com>.

```

FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_945/615529451.py in <cell line: 0>()
      11 CONFIDENCE_THRESHOLD = 0.5
      12
--> 13 model = YOLO(MODEL_PATH)
      14
      15 cap = cv2.VideoCapture(0) # 0 = default webcam; try 1 if you have multiple cameras

----- 9 frames -----
/usr/local/lib/python3.13/dist-packages/torch/serialization.py in __init__(self, name, mode)
    774 class _open_file(_opener[I0[bytes]]):
    775     def __init__(self, name: str | os.PathLike[str], mode: str) -> None:
--> 776         super().__init__(open(name, mode)) # noqa: SIM115
    777
    778     def __exit__(self, *args):

FileNotFoundError: [Errno 2] No such file or directory: 'best.pt'

```

```

1 # train.py
2 # Run this in Google Colab (Runtime > Change runtime type > GPU).
3 #
4 # Step 1: Install dependencies (Colab: run as a cell with a leading !)
5 # pip install ultralytics roboflow
6 #
7 # IMPORTANT: This downloads the dataset directly from Roboflow's servers into
8 # your Colab session. Nothing is uploaded from or processed on your own
9 # computer -- your laptop just needs a browser tab open. All the heavy work
10 # (download, unzip, training) happens on Google's cloud hardware.
11

```

```

12 import os
13 from roboflow import Roboflow
14 from ultralytics import YOLO
15 from google.colab import drive
16
17 # -----
18 # STEP 0: Mount Google Drive so we can safely save results at the end
19 # -----
20 drive.mount('/content/drive')
21
22 # -----
23 # STEP 1: Download the dataset directly into Colab (not through your PC)
24 # -----
25 rf = Roboflow(api_key="IN9sfHttwwoh80KDb0F2") # roboflow.com -> Settings -> API Key
26 project = rf.workspace("lexiwave-glqpu").project("construction-worker-9yqmx")
27 version = project.version(1)
28 dataset = version.download("yolov8")
29
30 DATA_ROOT = dataset.location # e.g. "construction-worker-9yqmx-1"
31
32 # -----
33 # STEP 2: Remove the 'boots' class from labels and re-index the rest
34 # -----
35 # Original order (alphabetical, per data.yaml): 0=boots, 1=gloves, 2=helmet, 3=human, 4=vest
36 # New order after dropping boots: 0=gloves, 1=helmet, 2=human, 3=vest
37 BOOTS_INDEX = 0
38 REMAP = {1: 0, 2: 1, 3: 2, 4: 3} # old_index -> new_index (boots/0 is dropped)
39
40 def clean_labels(split):
41     label_dir = os.path.join(DATA_ROOT, split, "labels")
42     for fname in os.listdir(label_dir):
43         if not fname.endswith(".txt"):
44             continue
45         path = os.path.join(label_dir, fname)
46         new_lines = []
47         with open(path, "r") as f:
48             for line in f:
49                 parts = line.strip().split()
50                 if not parts:
51                     continue
52                 old_cls = int(parts[0])
53                 if old_cls == BOOTS_INDEX:
54                     continue # drop boots annotations entirely
55                 new_cls = REMAP[old_cls]
56                 new_lines.append(" ".join([str(new_cls)] + parts[1:]))
57         with open(path, "w") as f:
58             f.write("\n".join(new_lines) + ("\n" if new_lines else ""))
59
60 for split in ["train", "valid", "test"]:
61     clean_labels(split)
62
63 # Write a corrected data.yaml with boots removed
64 new_yaml = f"""train: {DATA_ROOT}/train/images
65 val: {DATA_ROOT}/valid/images
66 test: {DATA_ROOT}/test/images
67
68 nc: 4
69 names: ['gloves', 'helmet', 'human', 'vest']
70 """
71 yaml_path = os.path.join(DATA_ROOT, "data_no_boots.yaml")
72 with open(yaml_path, "w") as f:
73     f.write(new_yaml)
74
75 print(f"Cleaned dataset ready. Using config: {yaml_path}")
76
77 # -----
78 # STEP 3: Train
79 # -----
80 model = YOLO("yolov8n.pt") # nano model: fastest to train, good baseline
81 # swap to "yolov8s.pt" for a bit more accuracy if training time allows
82
83 results = model.train(
84     data=yaml_path,
85     epochs=50,
86     imgsz=640,
87     batch=16,
88     patience=10, # early stopping if val loss plateaus

```

```

89     name="ppe_yolov8n",
90     project="runs/ppe"
91 )
92
93 # -----
94 # STEP 4: Evaluate
95 # -----
96 metrics = model.val()
97 print("mAP50-95:", metrics.box.map)
98 print("mAP50:", metrics.box.map50)
99 print("Per-class mAP50:", metrics.box.maps) # array aligned to class indices in data.yaml
100
101 # Confusion matrix and PR curves are auto-saved to runs/ppe/ppe_yolov8n/
102 # Look for confusion_matrix.png and results.png after training completes.
103
104 # -----
105 # STEP 5: Save your results to Google Drive (survives session disconnects)
106 # -----
107 import shutil
108 import glob
109
110 best_pt_candidates = glob.glob("runs/**/best.pt", recursive=True)
111 if not best_pt_candidates:
112     raise FileNotFoundError("Could not find best.pt -- check the runs/ folder structure printed above.")
113
114 best_pt_path = best_pt_candidates[0]
115 print("Found weights at:", best_pt_path)
116
117 DRIVE_SAVE_DIR = "/content/drive/MyDrive/ppe-detection-results"
118 os.makedirs(DRIVE_SAVE_DIR, exist_ok=True)
119
120 shutil.copy(best_pt_path, os.path.join(DRIVE_SAVE_DIR, "best.pt"))
121 confusion_matrix_path = best_pt_path.replace("weights/best.pt", "confusion_matrix.png")
122 results_png_path = best_pt_path.replace("weights/best.pt", "results.png")
123 if os.path.exists(confusion_matrix_path):
124     shutil.copy(confusion_matrix_path, os.path.join(DRIVE_SAVE_DIR, "confusion_matrix.png"))
125 if os.path.exists(results_png_path):
126     shutil.copy(results_png_path, os.path.join(DRIVE_SAVE_DIR, "results.png"))
127
128 print(f"Saved best.pt and evaluation images to your Google Drive at: {DRIVE_SAVE_DIR}")
129 print("Open drive.google.com on your computer and download from the 'ppe-detection-results' folder --")
130 print("this copy is safe even if this Colab session disconnects.")

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_reloading=True)

loading Roboflow workspace...

loading Roboflow project...

Downloading Dataset Version Zip in construction-worker-1 to yolov8:: 100%|██████████| 298347/298347 [00:09<00:00, 30

Extracting Dataset Version Zip to construction-worker-1 in yolov8:: 100%|██████████| 10008/10008 [00:01<00:00, 6446.

Cleaned dataset ready. Using config: /content/construction-worker-1/data_no_boots.yaml

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolov8n.pt': 100% |██████████|

Ultralytics 8.4.139 Python-3.13.15 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)

engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=16, bgr=0.0,

Downloading <https://ultralytics.com/assets/Arial.ttf> to '/root/.config/Ultralytics/Arial.ttf': 100% |██████████| 755

Overriding model.yaml nc=80 with nc=4

	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	7360	ultralytics.nn.modules.block.C2f	[32, 32, 1, True]
3	-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
4	-1	2	49664	ultralytics.nn.modules.block.C2f	[64, 64, 2, True]
5	-1	1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
6	-1	2	197632	ultralytics.nn.modules.block.C2f	[128, 128, 2, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8	-1	1	460288	ultralytics.nn.modules.block.C2f	[256, 256, 1, True]
9	-1	1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]
10	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
11	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
12	-1	1	148224	ultralytics.nn.modules.block.C2f	[384, 128, 1]
13	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
14	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
15	-1	1	37248	ultralytics.nn.modules.block.C2f	[192, 64, 1]
16	-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
17	[-1, 12]	1	0	ultralytics.nn.modules.conv.Concat	[1]
18	-1	1	123648	ultralytics.nn.modules.block.C2f	[192, 128, 1]
19	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
20	[-1, 9]	1	0	ultralytics.nn.modules.conv.Concat	[1]
21	-1	1	493056	ultralytics.nn.modules.block.C2f	[384, 256, 1]
22	[15, 18, 21]	1	752092	ultralytics.nn.modules.head.Detect	[4, 16, None, [64, 128, 256]]

Model summary: 130 layers, 3,011,628 parameters, 3,011,612 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights

Freezing layer 'model.22.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

AMP: downloading yolo26n.pt for AMP checks (one-time, not used for training)...

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt> to 'weights/yolo26n.pt': 100%

AMP: checks passed ✓

train: Fast image access ✓ (ping: 0.0±0.0 ms, read: 2166.7±823.3 MB/s, size: 63.5 KB)

train: Scanning /content/construction-worker-1/train/labels... 4368 images, 15 backgrounds, 0 corrupt: 100% _____

train: New cache created: /content/construction-worker-1/train/labels.cache

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weigh

val: Fast image access ✓ (ping: 0.0±0.0 ms, read: 572.5±422.4 MB/s, size: 66.6 KB)

val: Scanning /content/construction-worker-1/valid/labels... 420 images, 3 backgrounds, 0 corrupt: 100% _____

val: New cache created: /content/construction-worker-1/valid/labels.cache

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0'

optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 b

Plotting labels to /content/runs/detect/runs/ppe/ppe_yolov8n/labels.jpg...

Using 4368 train, 420 val images for fraction=1.0 at imgsz=640

Using 2 dataloader workers

1 Start coding or [generate](#) with AI.

1 !pip install roboflow ultralytics

Collecting roboflow

Downloading roboflow-1.4.2-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: ultralytics in /usr/local/lib/python3.13/dist-packages (8.4.139)

Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-packages (from roboflow) (2026.7.22)

Requirement already satisfied: idna>=3.7 in /usr/local/lib/python3.13/dist-packages (from roboflow) (3.19)

Requirement already satisfied: cycler in /usr/local/lib/python3.13/dist-packages (from roboflow) (0.12.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from roboflow) (1.5.0)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (from roboflow) (3.10.0)

Requirement already satisfied: numpy<2.4,>=1.18.5 in /usr/local/lib/python3.13/dist-packages (from roboflow) (2.1.3)

Requirement already satisfied: opencv-python-headless>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from roboflow) (4.11.0)

Requirement already satisfied: Pillow>=7.1.2 in /usr/local/lib/python3.13/dist-packages (from roboflow) (11.3.0)

Collecting pi-heif<2 (from roboflow)

Downloading pi_heif-1.4.0-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.3 kB)

Collecting pillow-avif-plugin<2 (from roboflow)

Downloading pillow_avif_plugin-1.6.0-cp313-cp313-manylinux_2_28_x86_64.whl.metadata (2.2 kB)

Requirement already satisfied: python-dateutil in /usr/local/lib/python3.13/dist-packages (from roboflow) (2.9.0.post0)

Requirement already satisfied: python-dotenv in /usr/local/lib/python3.13/dist-packages (from roboflow) (1.2.3)

Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (from roboflow) (2.32.4)

Requirement already satisfied: six in /usr/local/lib/python3.13/dist-packages (from roboflow) (1.17.0)

Requirement already satisfied: urllib3>=1.26.6 in /usr/local/lib/python3.13/dist-packages (from roboflow) (2.5.0)

Requirement already satisfied: tqdm>=4.41.0 in /usr/local/lib/python3.13/dist-packages (from roboflow) (4.67.3)

Requirement already satisfied: PyYAML>=5.3.1 in /usr/local/lib/python3.13/dist-packages (from roboflow) (6.0.3)

Requirement already satisfied: requests_toolbelt in /usr/local/lib/python3.13/dist-packages (from roboflow) (1.0.0)

Collecting filetype (from roboflow)

Downloading filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)

Collecting typer<0.26,>=0.12.0 (from roboflow)

Downloading typer-0.25.1-py3-none-any.whl.metadata (15 kB)

Requirement already satisfied: click>=8.0 in /usr/local/lib/python3.13/dist-packages (from roboflow) (8.5.0)

Requirement already satisfied: filelock>=3.16.1 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (3.32.0)

Requirement already satisfied: opencv-python<=4.13.0.90,>=4.7.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (4.11.0)

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.11.0+cu118)

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (0.15.2)

Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (5.9.5)

Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (1.35.2)

Requirement already satisfied: nvidia-ml-py>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (12.550.53)

Requirement already satisfied: ultralytics-thop>=2.1.6 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (2.1.1)

Requirement already satisfied: ultralytics-platform>=0.1.3 in /usr/local/lib/python3.13/dist-packages (from ultralytics) (0.1.3)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib->roboflow) (1.3.0)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib->roboflow) (4.55.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib->roboflow) (24.2)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib->roboflow) (3.2.0)

Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.13/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests->roboflow) (3.4.1)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (4.12.0)

Requirement already satisfied: setuptools>=82 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (75.1.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (1.13.3)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.4.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.1.4)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (2025.10.0)

Requirement already satisfied: cuda-toolkit>=12.8.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (12.8.1)

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (12.9.4)

Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (9.19.0.56)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (2.28.9)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.4.5)

Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (3.6.0)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (12.8.4.1)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.0 in /usr/local/lib/python3.13/dist-packages (from torch>=1.8.0->ultralytics) (12.8.0)

```
1 python3 infer_webcam.py
```

```
File "/tmp/ipykernel_945/3997975889.py", line 1
python3 infer_webcam.py
    ^
SyntaxError: invalid syntax
```

```
1 Start coding or generate with AI.
```